

Jinshi (Peter) Chen

Andlinger Center for Energy and the Environment, #016
86 Olden St., Princeton, NJ 08540
(607) 379-7985 • jinshic@princeton.edu
jinshichen.github.io

EDUCATION

Woods Hole Oceanographic Institution, Physical Oceanography Cambridge & Woods Hole, MA
Massachusetts Institute of Technology, Earth, Atmospheric, and Planetary Sciences Jun. 2019-Mar. 2025
Doctor of Philosophy
Advisor: Dr. Britt Raubenheimer & Dr. Steve Elgar
Dissertation: *Cross-shore Transformation of Breaking Random Waves in the Surfzone*

Cornell University, Physics Ithaca, NY
Bachelor of Arts, Magna Cum Laude Aug. 2015-May 2019
Overall GPA: 4.08/4.00 • Major GPA: 4.14/4.00

POSITIONS

Princeton University, Andlinger Center for Energy and the Environment Princeton, NJ
Distinguished Postdoctoral Fellow Apr. 2025-Present

Woods Hole Oceanographic Institution, Applied Ocean Physics & Engineering Woods Hole, MA
Guest Investigator Apr. 2025-Present

RESEARCH AREAS & INTERESTS

• Nearshore processes & circulation • Air-sea interaction • Wave breaking & turbulence • Scientific modeling
• Coastal resilience & climate mitigation • Marine renewable energy

PUBLICATIONS

• Chen, J., Scapin, N., & Deike, L. Momentum and Energy Exchange in Misaligned Wind-Wave Conditions. *In prep*
• Chen, J., Raubenheimer, B., & Elgar, S. Depth-resolved Surfzone Wave and Roller Transformation. *Under revision*.
• Chen, J., Raubenheimer, B., & Elgar, S. (2025). Anisotropy of Surfzone Turbulence. *Journal of Physical Oceanography*.
• Chen, J., Raubenheimer, B., & Elgar, S. (2024). Wave and Roller Transformation over Barred Bathymetry, *Journal of Geophysical Research: Oceans*, 129, e2023JC020413

INVITED PRESENTATIONS

• *A Peek into Misaligned Oceanic Wind-Wave Interactions with High-Fidelity Models*, New Light: Rising Stars in Energy and the Environment, Princeton University, Jun. 2026
• *Cross-shore Transformation of Breaking Random Waves in the Surfzone*, Shanghai Jiao Tong University, Apr. 2025
• *Cross Surfzone Transport Dynamics by Random Breaking Waves*, Sack Lunch Seminar, MIT, Dec. 2024
• *Cross Surfzone Transport Dynamics by Random Breaking Waves*, Center for Coastal Studies seminar, Scripps Institution of Oceanography, Oct. 2024
• *Simulations and Observations of Surfzone Waves and Undertow*. Coastal Ocean Fluid Dynamics Laboratory (COFDL) seminar, WHOI, Aug. 2021

CONFERENCE PRESENTATIONS

- Chen, J., Scapin, N., & Deike, L. (2026, May). *A DNS Study of Momentum and Energy Exchange in Misaligned Wind-Wave Conditions*. Presented at 2026 International Ocean Vector Winds Science Team Meeting.
- Chen, J., Scapin, N., Wu, J., & Deike, L. (2025, November). *Momentum and Energy Exchange in Misaligned Wind-Wave Conditions*. Presented at 2025 Annual Meeting of the APS Division of Fluid Dynamics.
- Chen, J., Raubenheimer, B., Elgar, S., & Tsai, B. (2024, December). *Simulations of Depth Resolved Cross-shore Momentum Transfer in the Surf Zone*. Poster presented at 2024 AGU Fall Meeting.
- Chen, J., Trowbridge, J., Raubenheimer, B., & Elgar, S. (2024, February). *Observations of Surfzone Turbulence Anisotropy*. Presented at 2024 Ocean Sciences Meeting.
- Chen, J., Raubenheimer, B., & Elgar, S. (2023, August). *Tidal Effects of Cross-shore Roller Transformation over Barred Bathymetries*. Presented at 2023 Young Coastal Scientists and Engineers Conference-Americas.
- Chen, J., Raubenheimer, B., & Elgar, S. (2022, December). *Cross-shore Roller Transformation over Barred Bathymetries*. Presented at 2022 AGU Fall Meeting.
- Chen, J., Raubenheimer, B., & Elgar, S. (2018, November). *Surfzone Setup and Alongshore Currents During Hurricane Matthew*. Poster presented at 2018 Annual Meeting of the APS Division of Fluid Dynamics.

HONORS & AWARDS

- MathWorks Fellowship Aug. 2023-Jun. 2024
- Merrill Presidential Scholar May 2019
- American Physical Society Division of Fluid Dynamics student travel grant Sept. 2018
- Woods Hole Oceanographic Institution Summer Student Fellowship Jun. 2018-Aug. 2018
- National Marine Figure of the Year 2013 by State Oceanic Administration (SOA), China Jun. 2014

RESEARCH GRANTS

- MathWorks Fellowship (~\$100k) Aug. 2023-Jun. 2024
- Ocean Ventures Fund (~\$10k) Jun. 2023
- PADI Foundation Grant (~\$8k) Apr. 2021

RESEARCH EXPERIENCE

- Princeton University, Andlinger Center for Energy and the Environment** Princeton, NJ
- Postdoctoral Fellow Collaborating PIs: Prof. Luc Deike & Prof. Michael Mueller** Apr. 2025-Present
- Simulate misaligned wind-wave interactions using two-phase DNS *Basilisk* framework.
 - Derive parametrization of momentum and energy transfer under misaligned wind-wave conditions.
 - Scale up the modeled results in offshore wind turbine models.

- MIT/WHOI Joint Program** Cambridge, MA & Woods Hole, MA
- Graduate Student Advisor: Prof. Glenn Flierl** Dec. 2020-Apr. 2022
- Simulated random vortex generation and advection over a slope using the *Dedalus* framework.
 - Derived dimensionless relation between vortex properties (e.g, energy, size) and slope & bathymetry properties.
 - Explored possible critical transitions of vortex advection using a simplified vortex dipole advection model.

- MIT/WHOI Joint Program** Cambridge, MA & Woods Hole, MA
- Graduate Student Advisors: Dr. Britt Raubenheimer & Dr. Steve Elgar** Jun. 2019-Mar. 2025
- Numerically investigated the cross-shore wave roller transformation and vertical structure of undertow with random waves and field-based bathymetry using *OpenFOAM*.
 - Investigated the surfzone turbulence structure and anisotropy with its relation to wave and mean current conditions using field observation.

FIELD WORK

- SINKEX, Field Research Facility, Duck, NC** Sept. 2023-Oct. 2023
- Deployed vertically aligned ADVs to measure turbulence (Funded by a WHOI OVF Award).
 - Assisted mounting field cameras to collect surf zone foam images.

BIASEX, Field Research Facility, Duck, NC

Sept. 2022

- Deployed ADVs and ADCPs in surf zone.
- Assisted with mounting field cameras to collect surf zone foam images.
- Conducted drone operations for remote sensing on surface currents.

DUNEX, Field Research Facility, Duck, NC

Aug. 2021-Oct. 2021

- Deployed ADV, ADCP and pressure sensors in surf zone to measure the vertical structure of undertow (Funded by PADI Foundation Award).
- Assisted mounting field cameras to collect surf zone foam images.
- Provided daily maintenance for in-situ sensors.

PROFESSIONAL ACTIVITIES

- **Chair & Convener**, Nearshore Processes, 2024 AGU Fall Meeting, Washington, D.C.

SPECIALIZED SKILLS

Software: MATLAB • Mathematica • FLUENT • LabVIEW**Programming languages:** Python • C/C++ • Julia**Numerical models:** OpenFOAM • Dedalus • Basilisk**Laboratory:** Particle Image Velocimetry (PIV) • Laser Induced Fluorescence (LIF) • Raman spectroscopy**Field Work:** Acoustic Doppler Velocimetry (ADV) • Acoustic Doppler Current Profiler (ADCP) • CTD**MENTORING & TEACHING**

Guest Instructor, Princeton University

Princeton, NJ

- Taught nearshore hydrodynamics and sediment dynamics in MAE330/CEE330: Ocean Waves

Oct. 2025

Math Review Instructor, MIT-WHOI Joint Program

Woods Hole, MA

- Taught a signal processing session for incoming MIT-WHOI Joint Program students.

Aug. 2024

Grad Teaching Development Tracks, MIT

Cambridge, MA

- Develop teaching skills through interactive learning modules and literature.

Nov. 2022-Feb. 2024

Peer Advisor, Cornell University

Ithaca, NY

- Mentored six first-year students on physics course selections and finding research projects.

Aug. 2017-Dec. 2017

Undergraduate Teaching Assistant, Cornell University

Ithaca, NY

- Held discussion sessions for PHYS 1112: Mechanics & Heat.

Feb. 2016-May 2016

AFFILIATIONS

- American Geophysical Union

Dec. 2022-Present

- Phi Beta Kappa Honor Society

Mar. 2019-Present

- American Physical Society

Sept. 2018-Present

ADDITIONAL EXPERIENCE

- **Guest Student**, Woods Hole Oceanographic Institution, Woods Hole, MA

Jul. 2016-Aug. 2016

- **Leading Student Researcher on Cyanobacteria**, Tsinghua University, Beijing, China

Sept. 2013-Aug. 2014

CERTIFICATION

- **MIT Grant Writing Certificate**

Jun. 2024

- **MIT Research Mentoring Certificate**

Apr. 2024

- **MIT Graduate Teaching Certificate**

Apr. 2024

- **Scientific Diver**, American Academy of Underwater Sciences

Jul. 2021